



Search and Rescue drone

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Problem Statement

How to design a drone fleet capable of effectively locating, detecting, and alerting to the presence of a person in distress?



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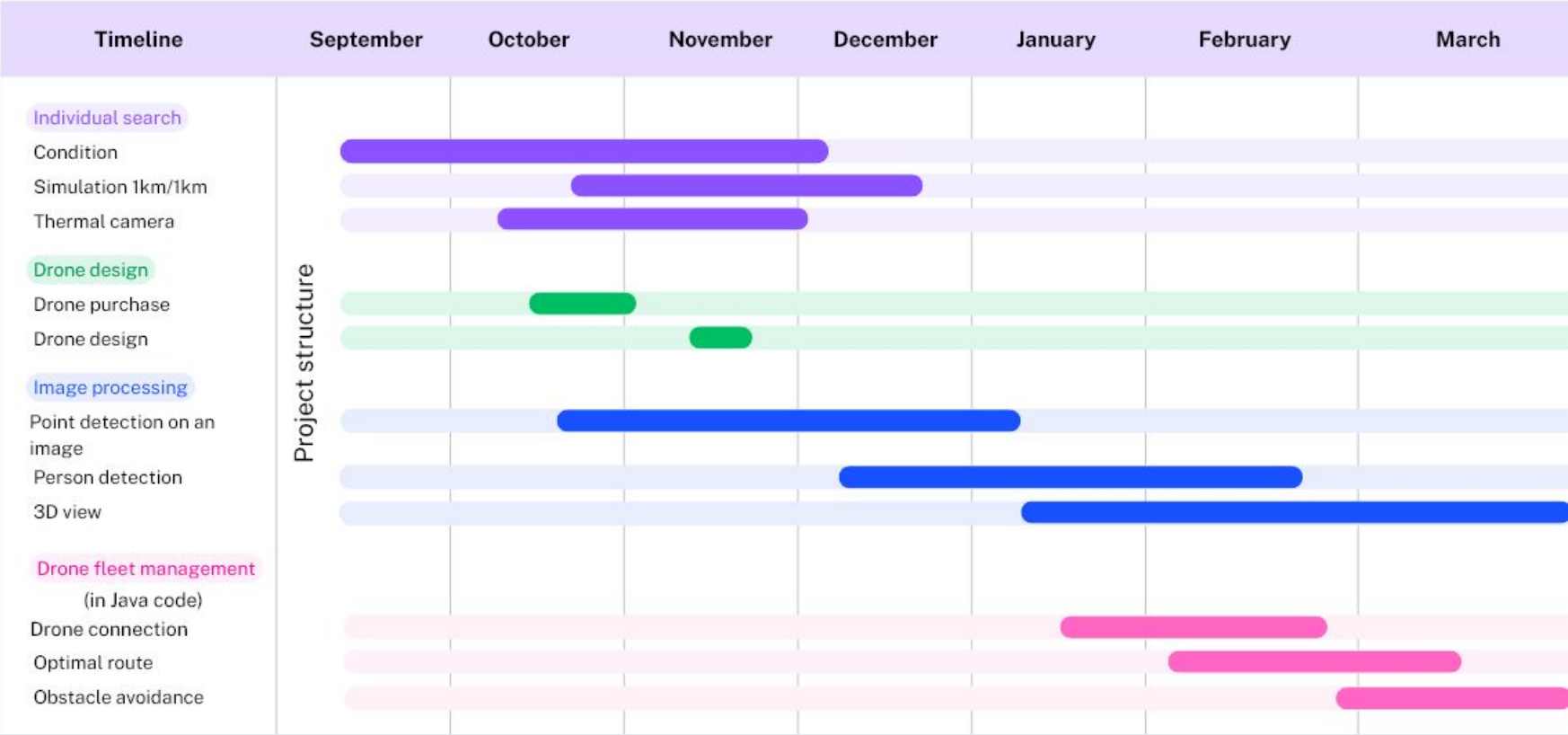
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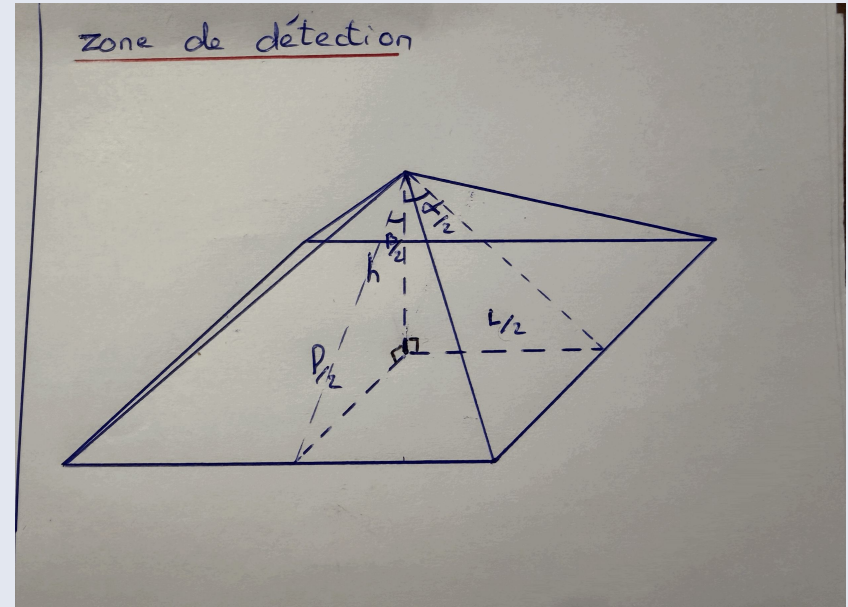
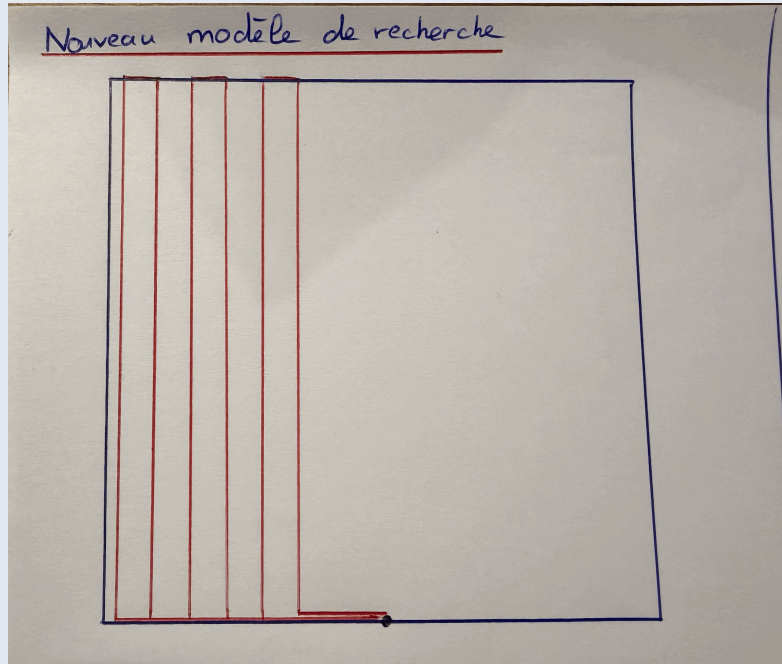
Gantt Chart





01 Human Search

Search for individuals on a flat 1 km² plot



→ this is our most realistic and optimal result

Search for individuals on a flat 1 km² plot

Height (m)	Length (m)	Width (m)	Nb of passes	distance (m)	distance covered by one drone (m)	Nb of drones
2	1,11	0,81	949	949916	6005	159
3	1,66	1,22	633	633610	6008	106
4	2,22	1,63	474	475456	6011	80
5	2,77	2,03	380	380564	6013	64
6	3,33	2,44	316	317302	6016	53
7	3,88	2,85	271	272115	6018	46
8	4,44	3,26	237	238225	6021	40
9	4.99	3.66	211	211866	6024	36
10	5,55	4,07	190	190778	6026	32

For H=10 : 22 min → 7,26 km



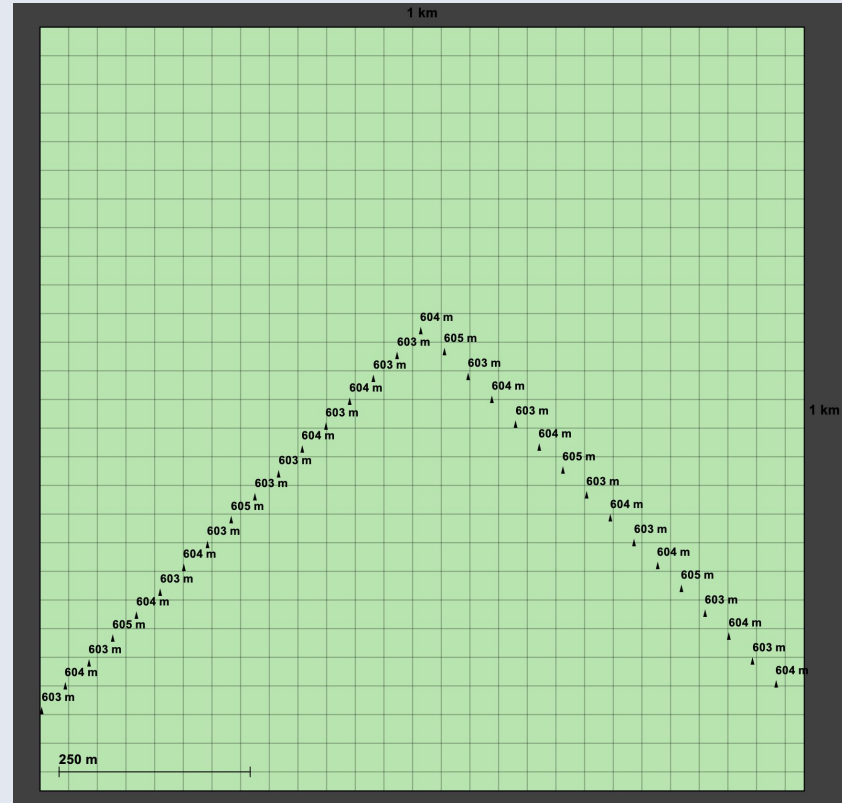
02

Drone fleet management

→ With java code

Java code 1

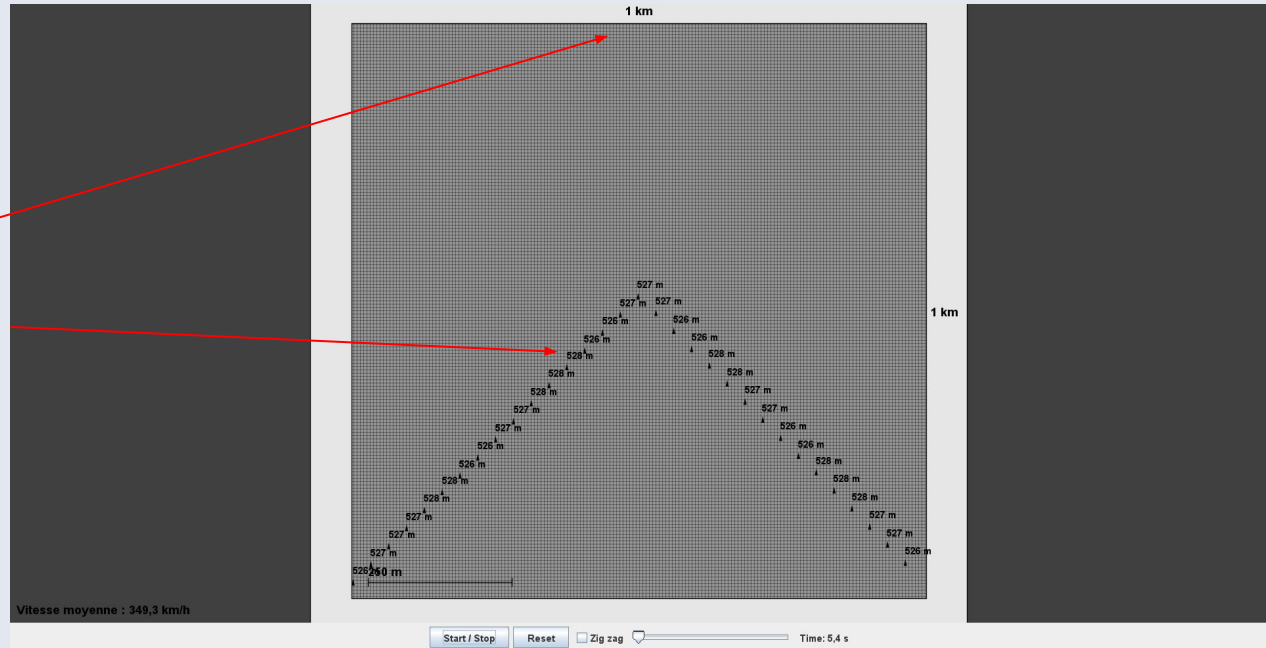
- Pattern
- Informations
- Grid



Java code 2

Step completed

- Square 1 km²
- Distance travelled by each drone
- Drone speed



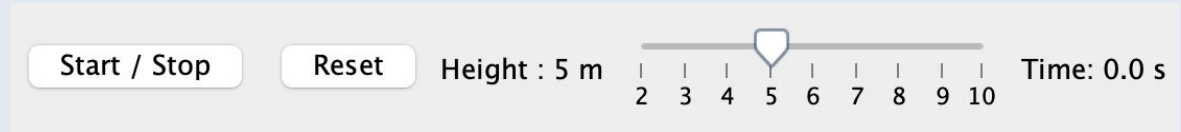
Java Code 2

- Colouring of the area already scanned
- Parameters of the simulation



Java Code 3

- Height control



A horizontal control bar with a light beige background. It contains two buttons: 'Start / Stop' and 'Reset'. To the right of the buttons is the text 'Height : 5 m'. Further right is a horizontal slider with a range from 2 to 10, with tick marks at every integer. A white triangular slider handle is positioned at the value 5. To the right of the slider is the text 'Time: 0.0 s'.

Important settings

- Number of drones
- Speed
- Time
- Distance

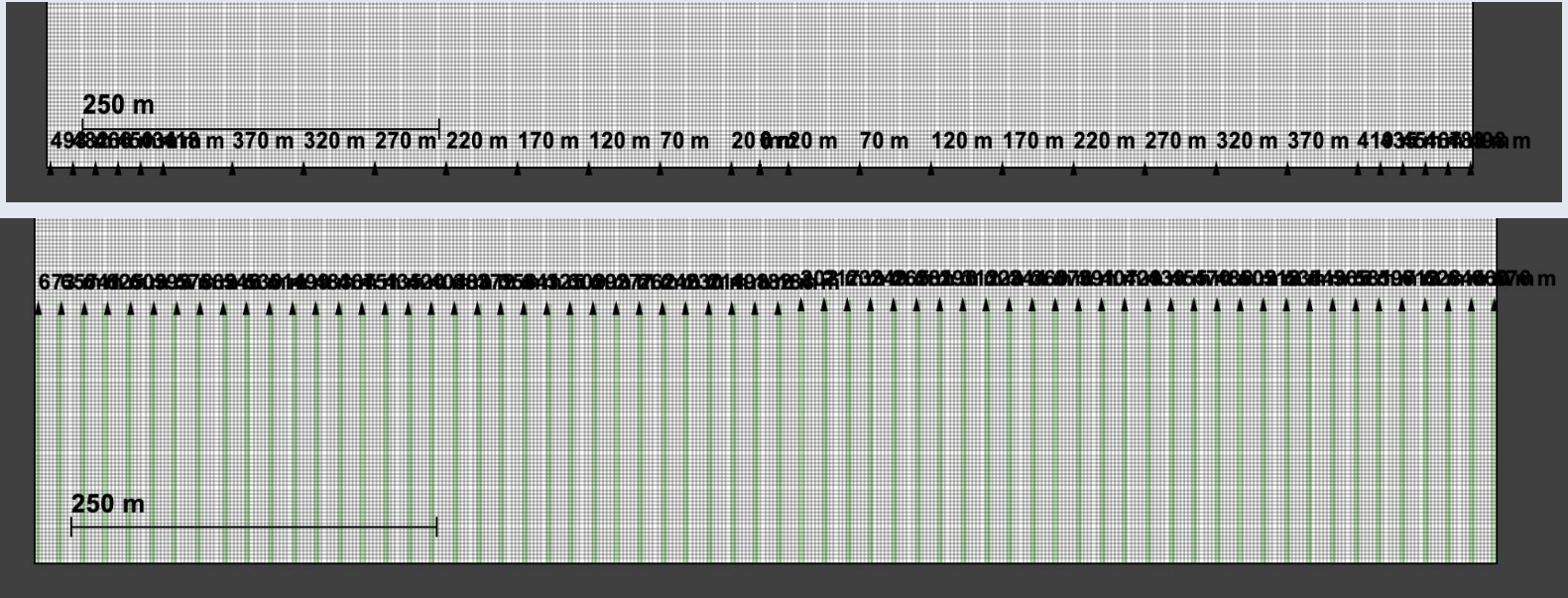
Number of drones : 32

Speed : 20,0 km/h

Time : 2 min 10 s

Distance (min / max) : 235 m / 724 m

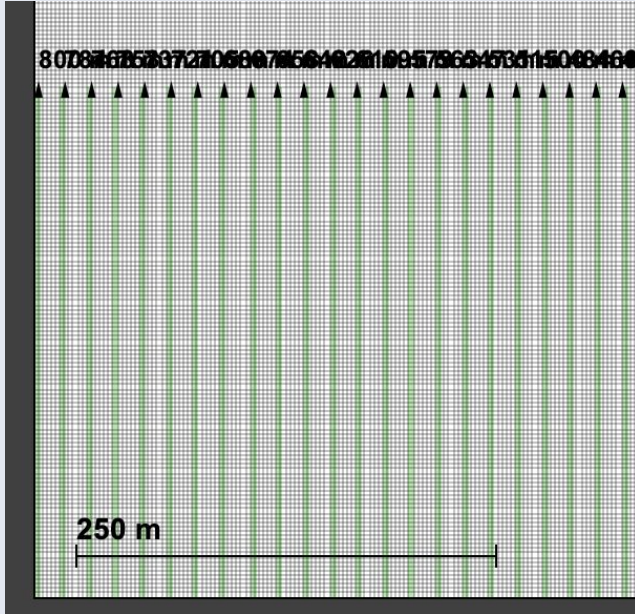
Java Code 3



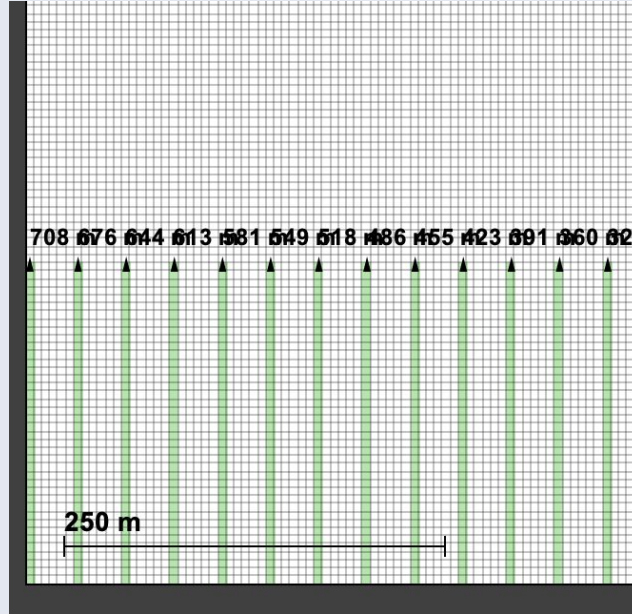
- Drone deployment time
- No collision

$$t_{deployment} = 1,134s \rightarrow 5,76s$$

Java Code 3



Height = 5 m

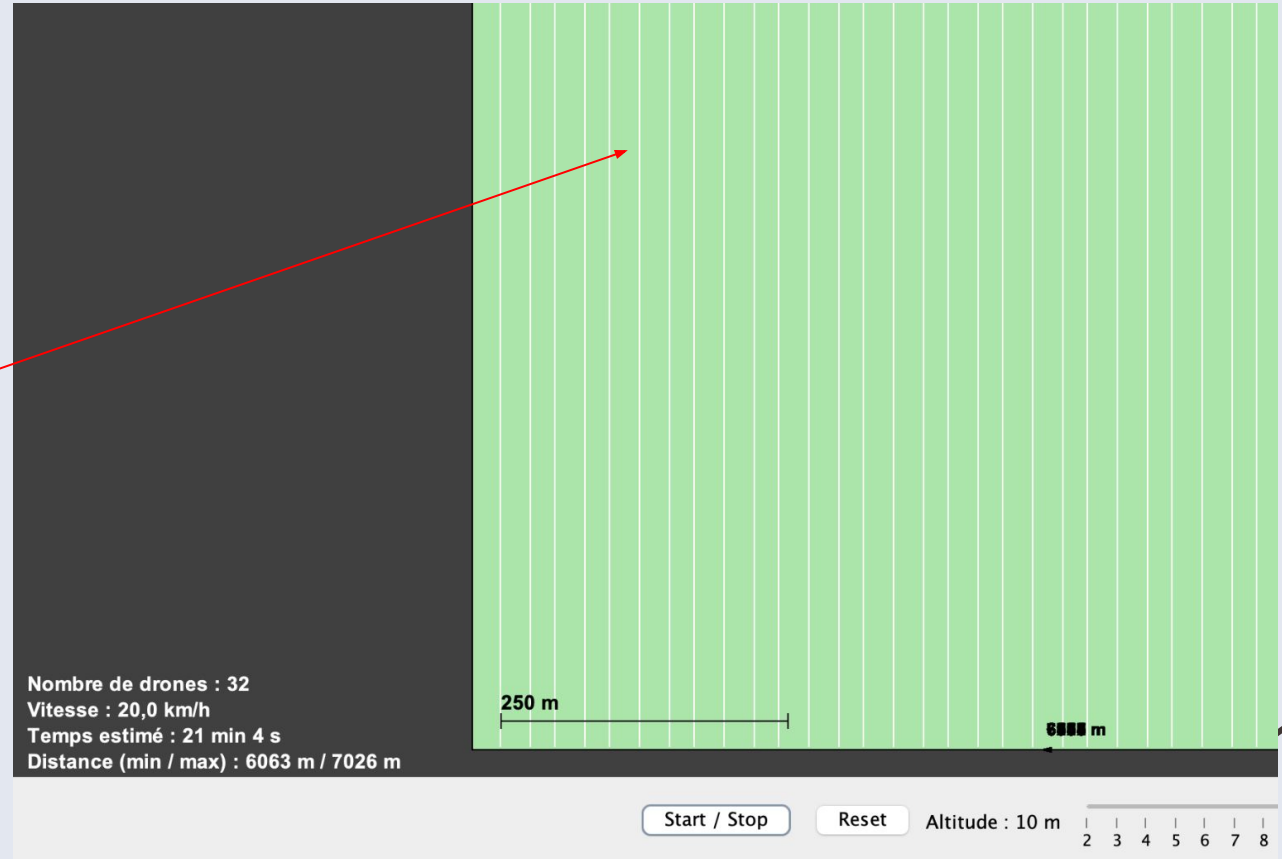


Height = 10 m

Adjustment of the grid according to the height of the drone

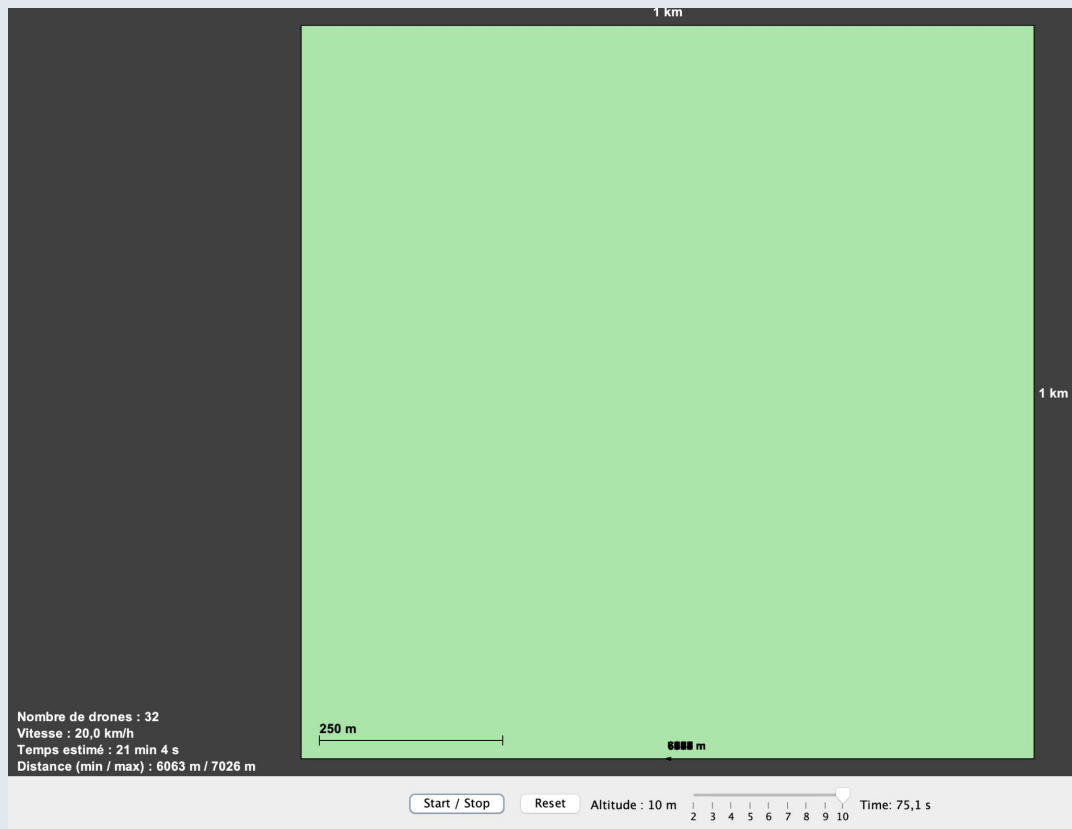
Java Code 4

- Remove the grid
- Problem with unexplored areas



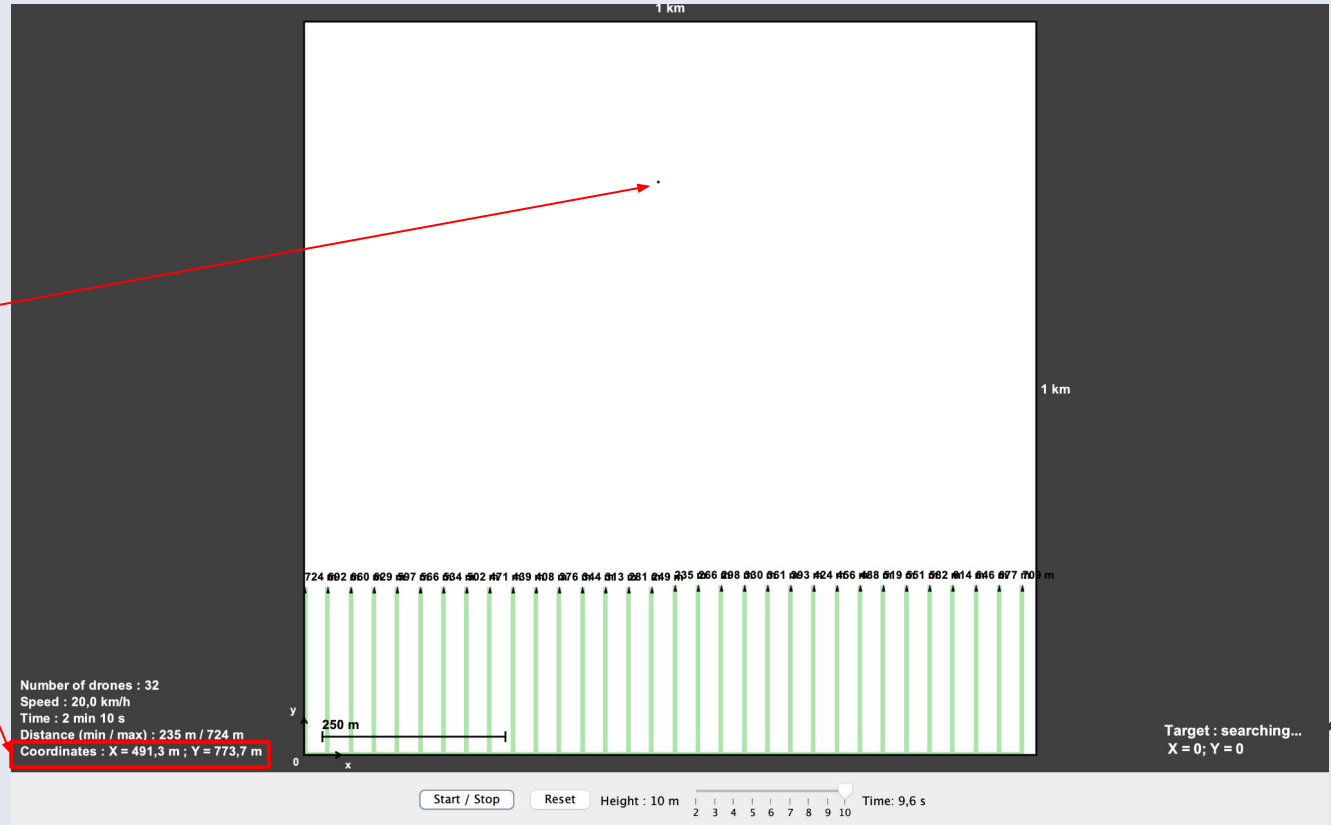
Java Code 4

We resolve the issue to ensure that the entire area is covered



Java Code 5

- Target
- Coordinates



Java Code 5

Number of drones : 32

Speed : 20,0 km/h

Time : 2 min 10 s

Distance (min / max) : 235 m / 724 m

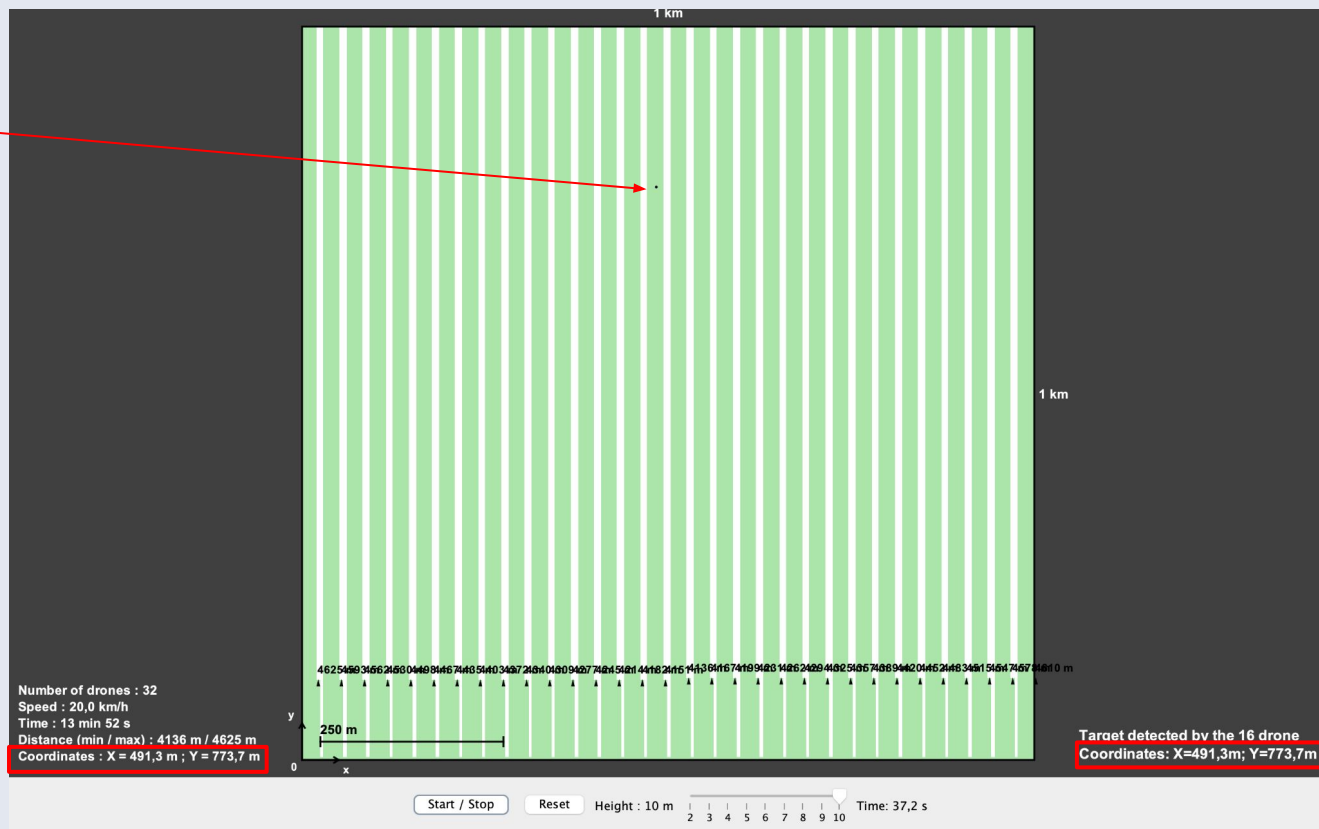
Coordinates : X = 491,3 m ; Y = 773,7 m



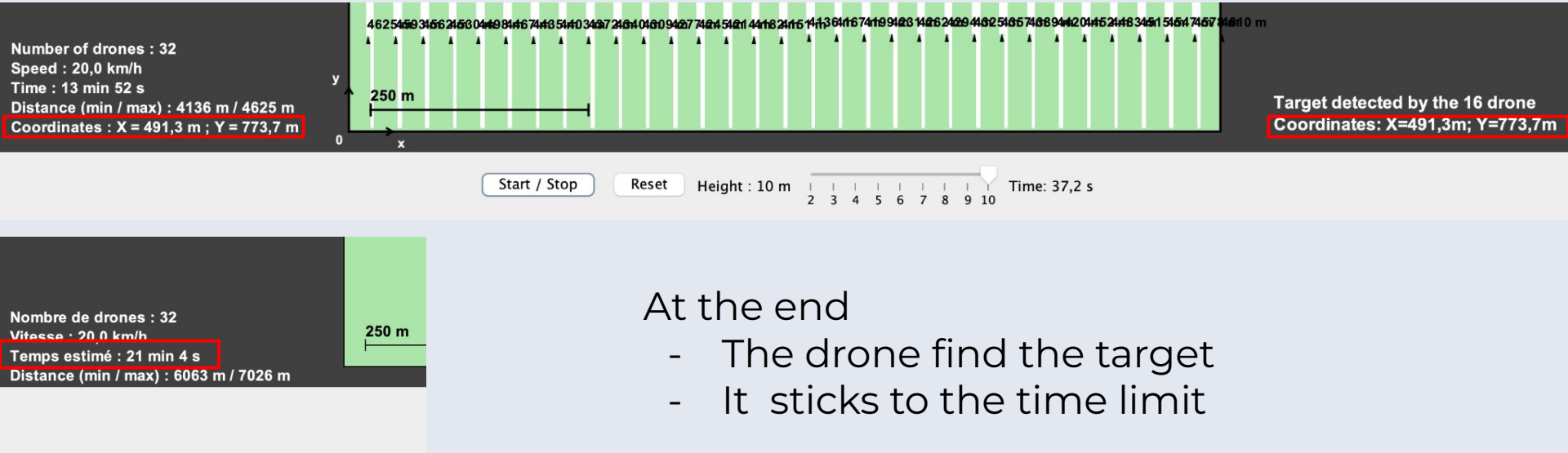
Start / Stop

Java Code 5

The target
is found



Java Code 5



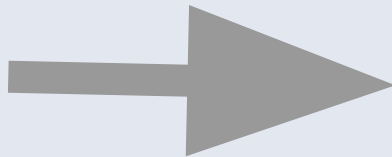
At the end

- The drone find the target
- It sticks to the time limit

Java code

Variable parameter :

- Shape of the research area
- Square dimension
- Number of drones
- Speed
- Time
- Height

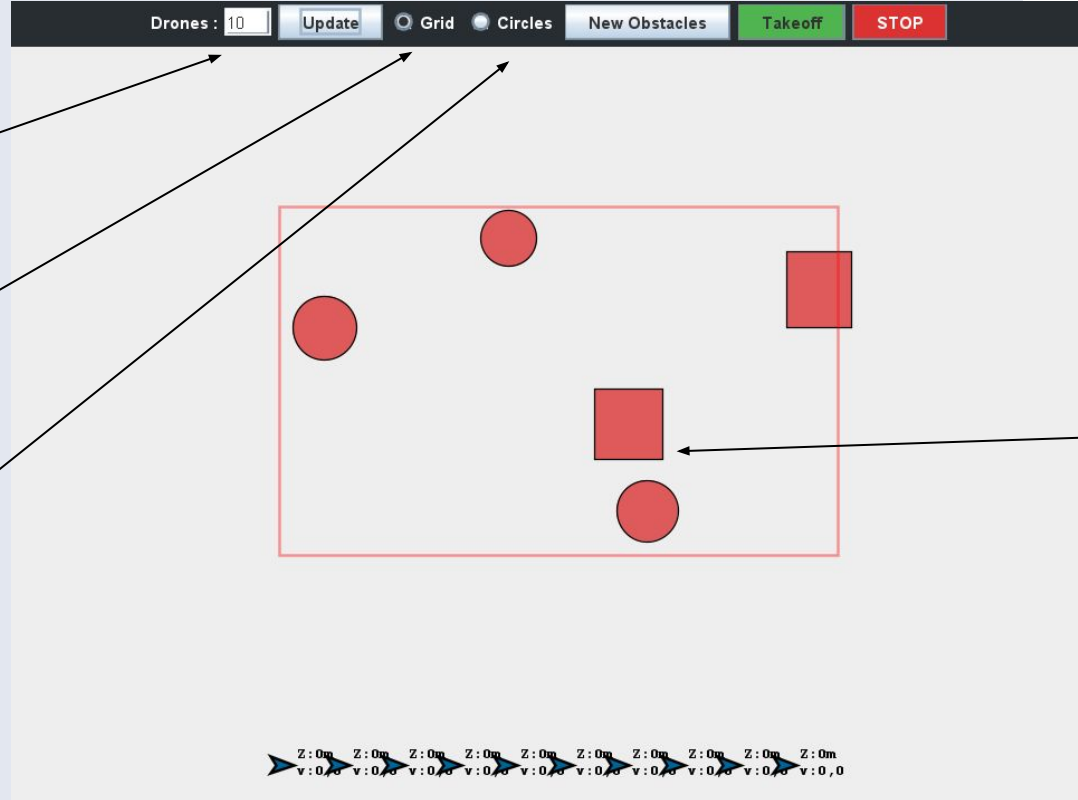


We can therefore test many different solutions to find the most optimal one but we haven't had time yet.

With this interface, we were able to consolidate all our previous calculations.

Java Code : obstacle avoidance

The interface

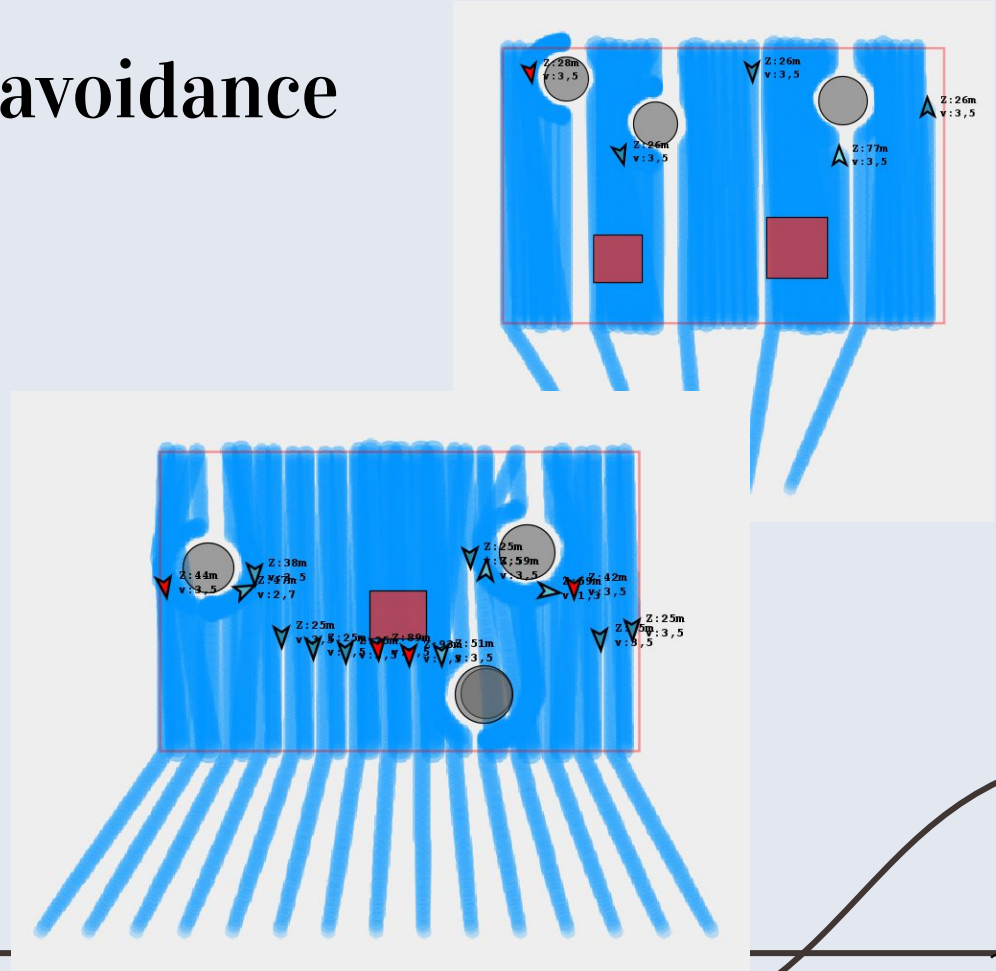


Java Code : obstacle avoidance

What does the code do?

THE GOAL

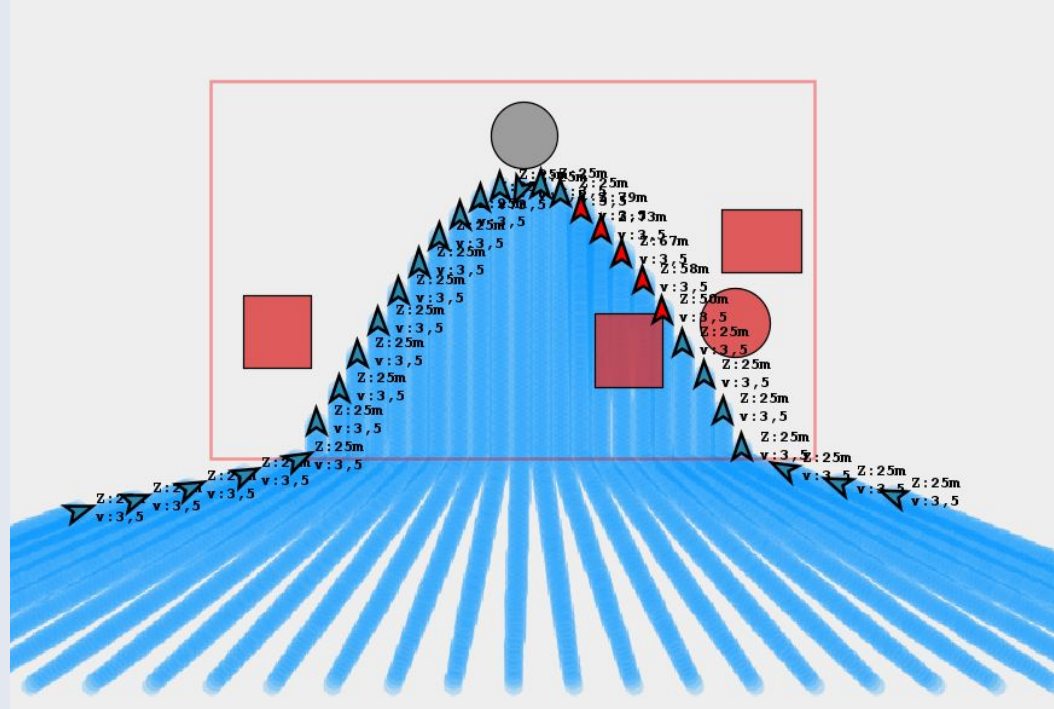
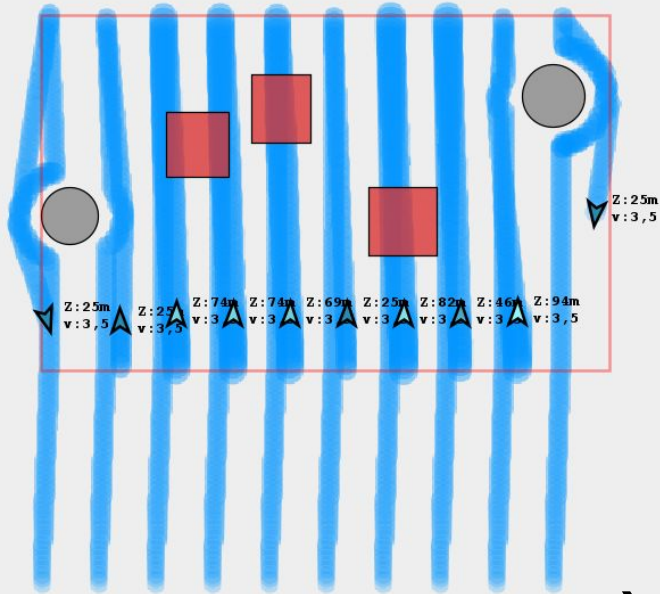
- Avoid obstacle :
 - A circle : go around
 - A rectangle : gain altitude
- red drone + height is changed



Java Code : obstacle avoidance

30 drones

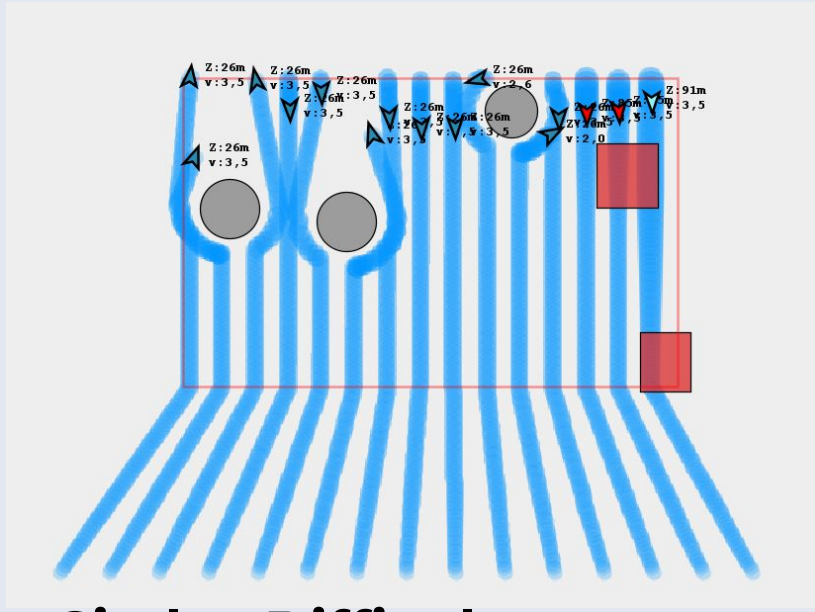
10 drones



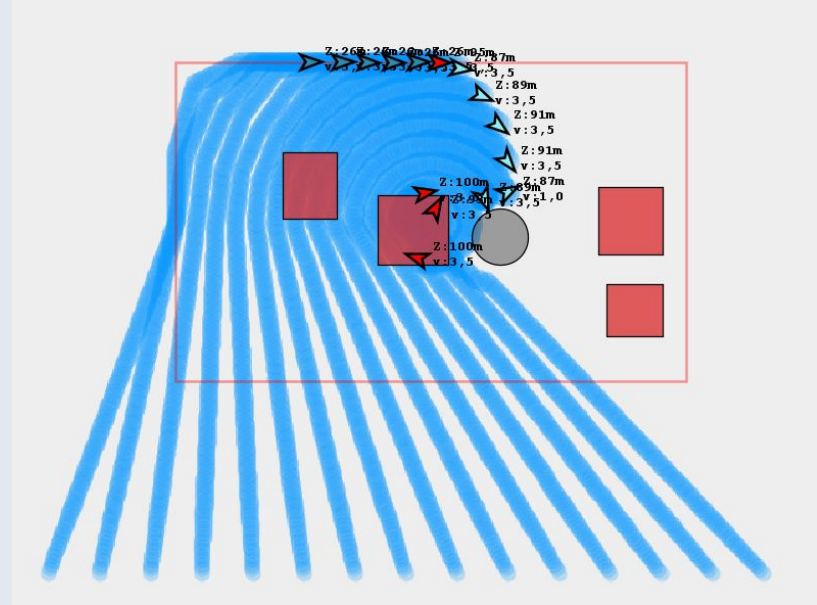
→ Fewer trips

Java Code : obstacle avoidance

Grid



Circle



→ Circle : Difficult to manage the overlapping between drones



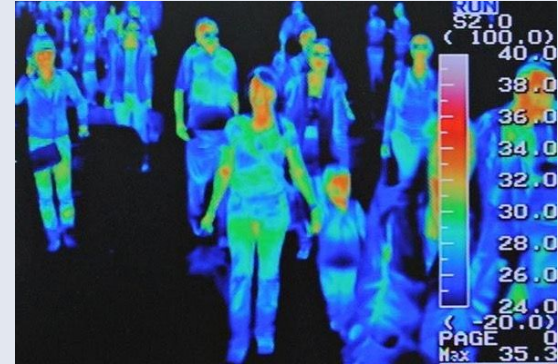
03

Machine Learning

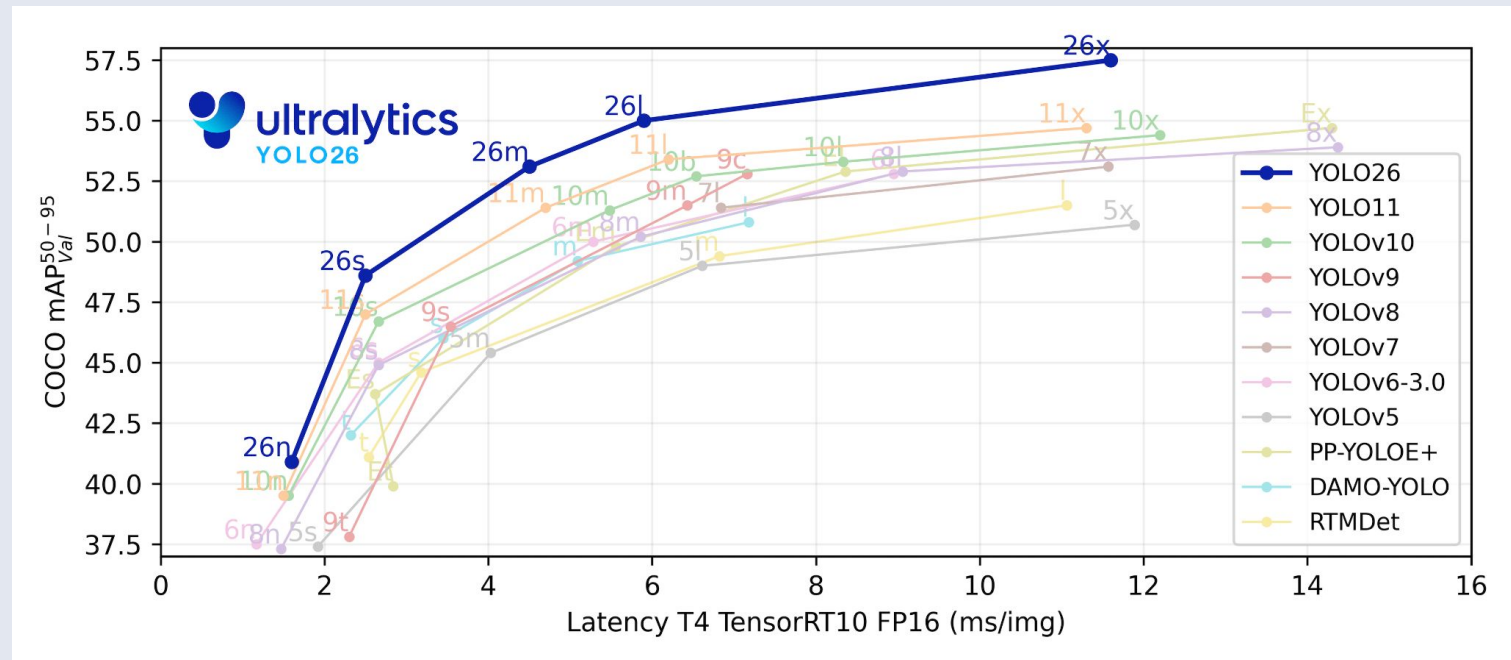
Regular or Thermal Camera



OR



Yolo Model



YOLO benchmark on 2 datasets

BENCHMARK RESULTS									
Model	F1	Score	Recall	Precision	Speed	TP	FP	FN	
YOLO26N.PT	36.5	26.3%	59.3%	15.2	img/s	35	24	98	
YOLO26S.PT	51.6	41.4%	68.7%	21.0	img/s	55	25	78	
YOLO26M.PT	56.2	47.4%	69.2%	20.0	img/s	63	28	70	
YOLO26L.PT	64.4	56.4%	75.0%	15.6	img/s	75	25	58	

SARD

BENCHMARK RESULTS									
Model	F1	Score	Recall	Precision	Speed	TP	FP	FN	
YOLO26N.PT	46.6	45.1%	48.1%	0.4	img/s	102	110	124	
YOLO26S.PT	53.3	67.3%	44.2%	0.4	img/s	152	192	74	
YOLO26M.PT	57.2	68.1%	49.4%	0.4	img/s	154	158	72	
YOLO26L.PT	58.3	71.2%	49.4%	0.4	img/s	161	165	65	

HERIDAL

Used Datasets and Retraining



SARD

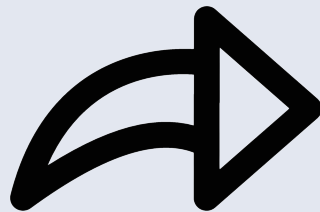


HERIDAL

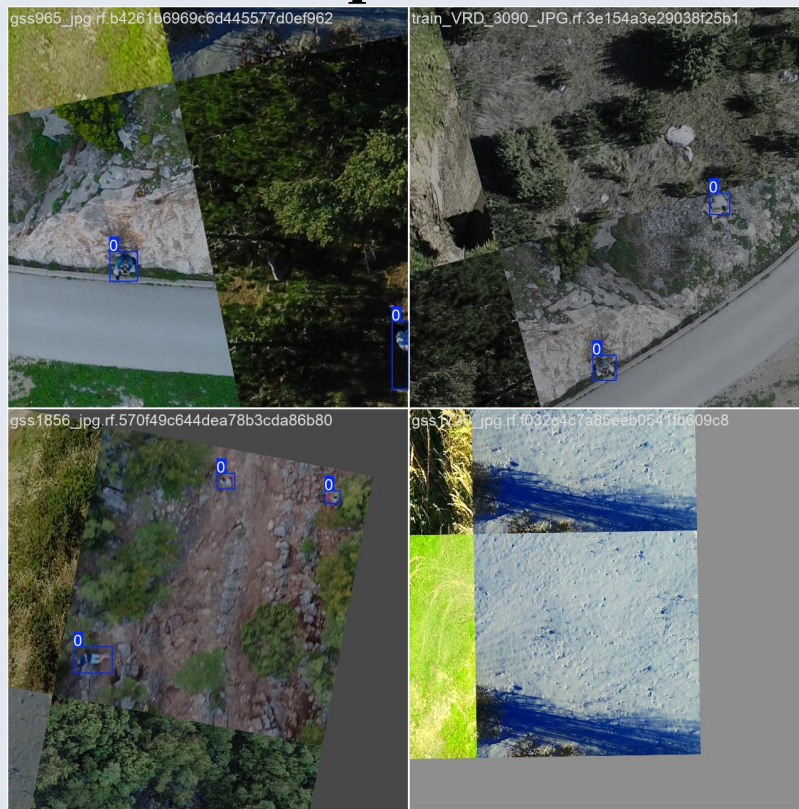
The HERIDAL's 4K Image Challenge



Processing large photos



Training batch example



Results after 40 epochs (10 hours)

BENCHMARK RESULTS									
Model	F1	Score	Recall	Precision	Speed	TP	FP	FN	
production_run_.../last.pt		95.2	91.6%	99.2%	15.2 img/s	120	1	11	
YOLO26M.PT		55.7	46.6%	69.3%	19.5 img/s	61	27	70	

SARD

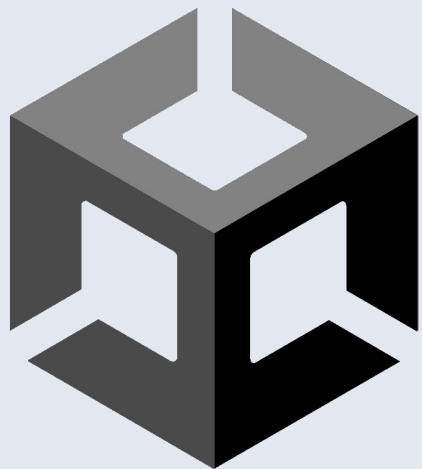
BENCHMARK RESULTS									
Model	F1	Score	Recall	Precision	Speed	TP	FP	FN	
production_run_.../last.pt		86.7	95.1%	79.6%	0.4 img/s	234	60	12	
YOLO26M.PT		50.6	63.0%	42.2%	0.4 img/s	155	212	91	

HERIDAL

04

Unity Simulation

Why Unity ?

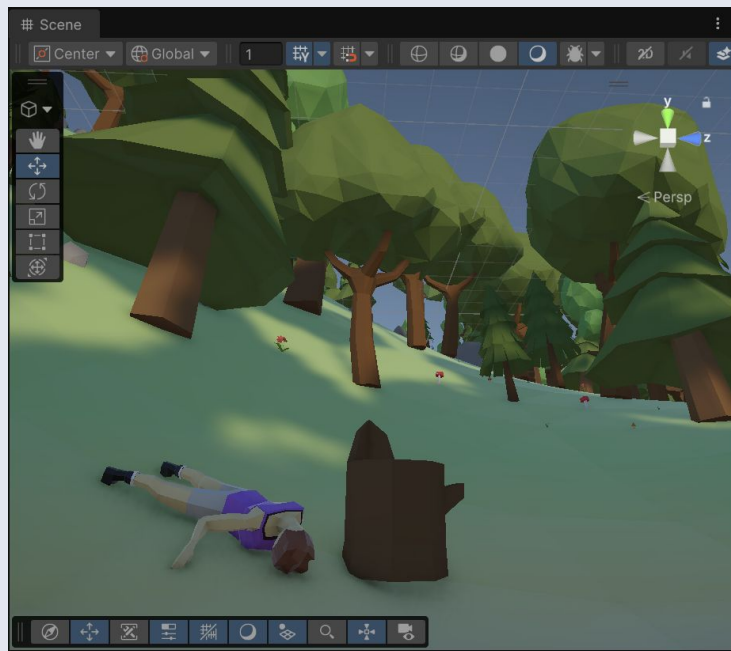


Unity

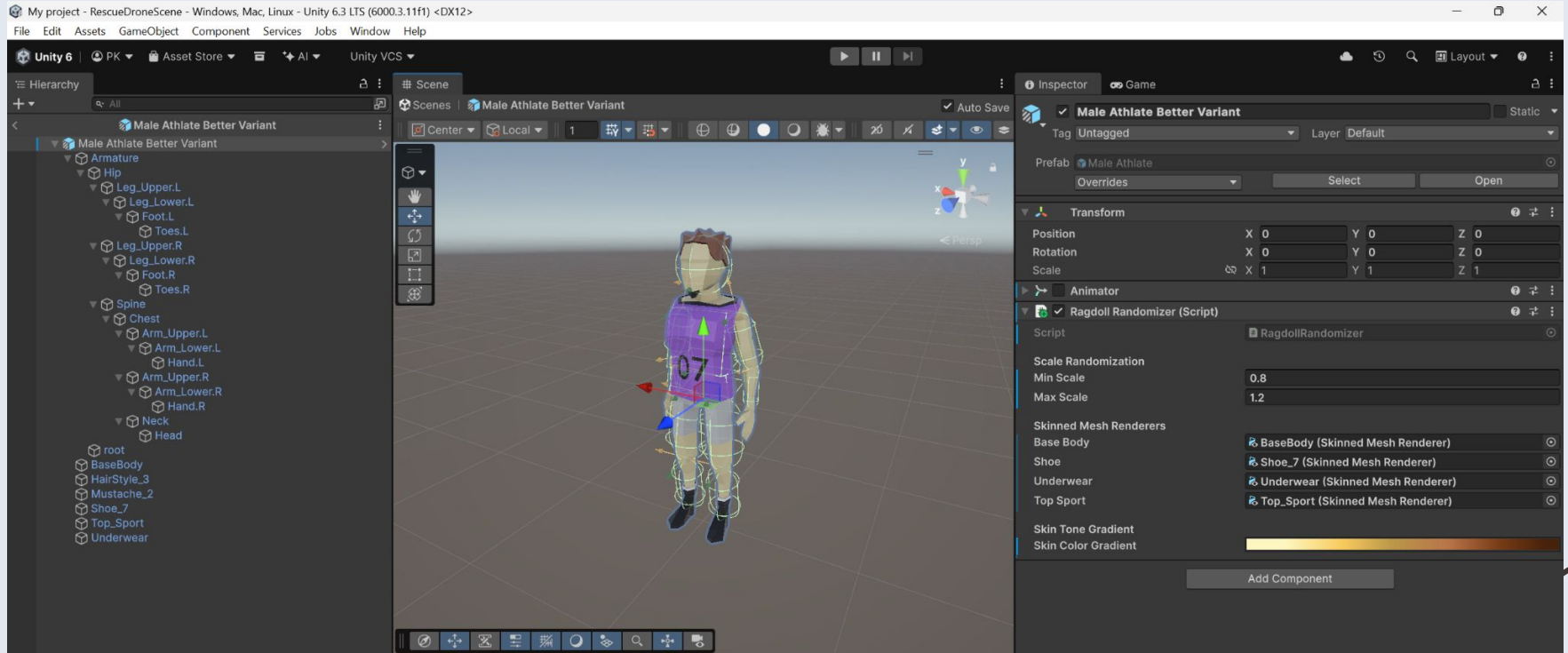
Our World



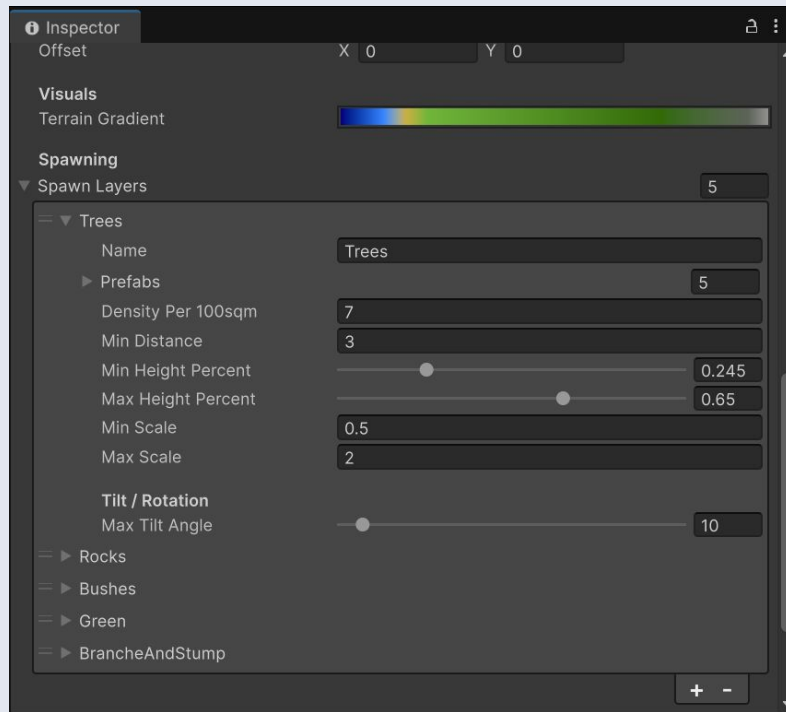
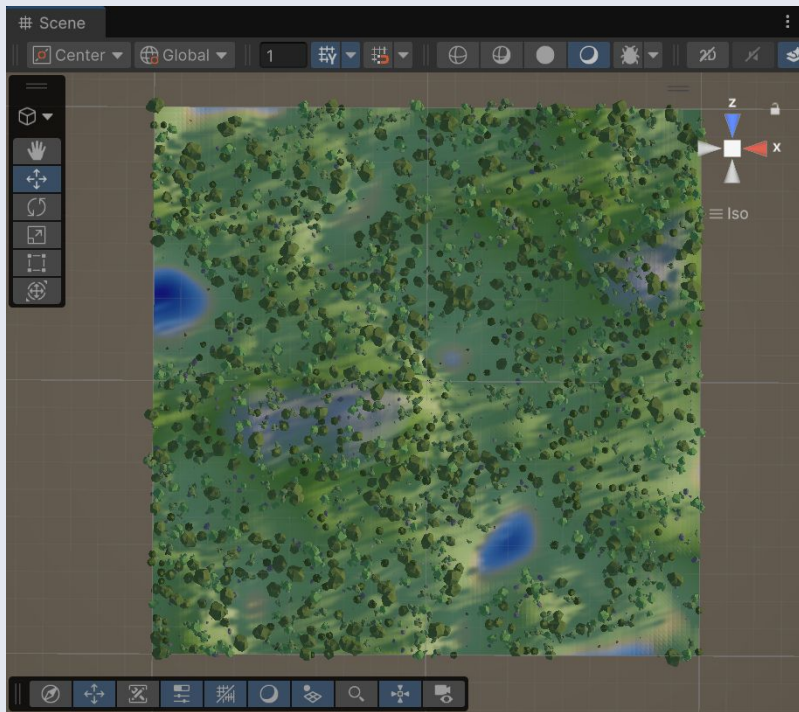
Low Polygon



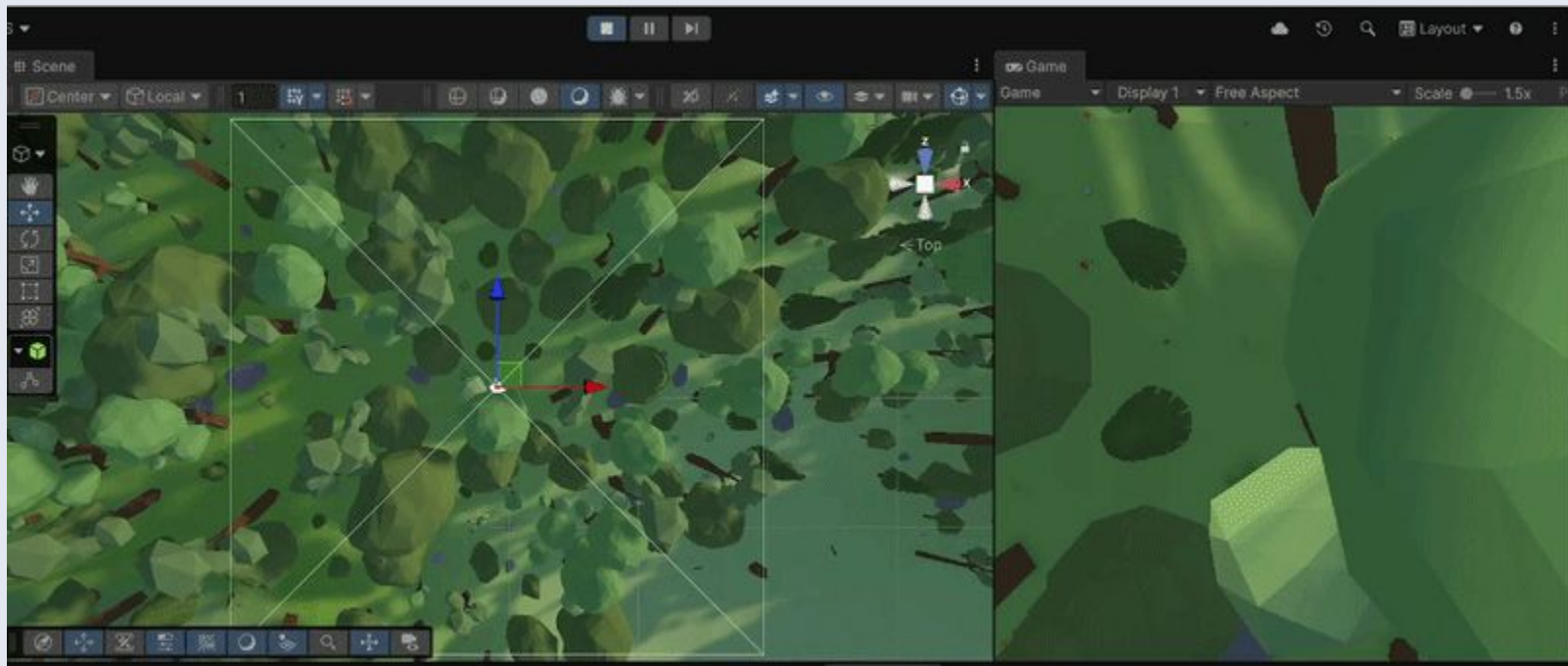
Character randomization



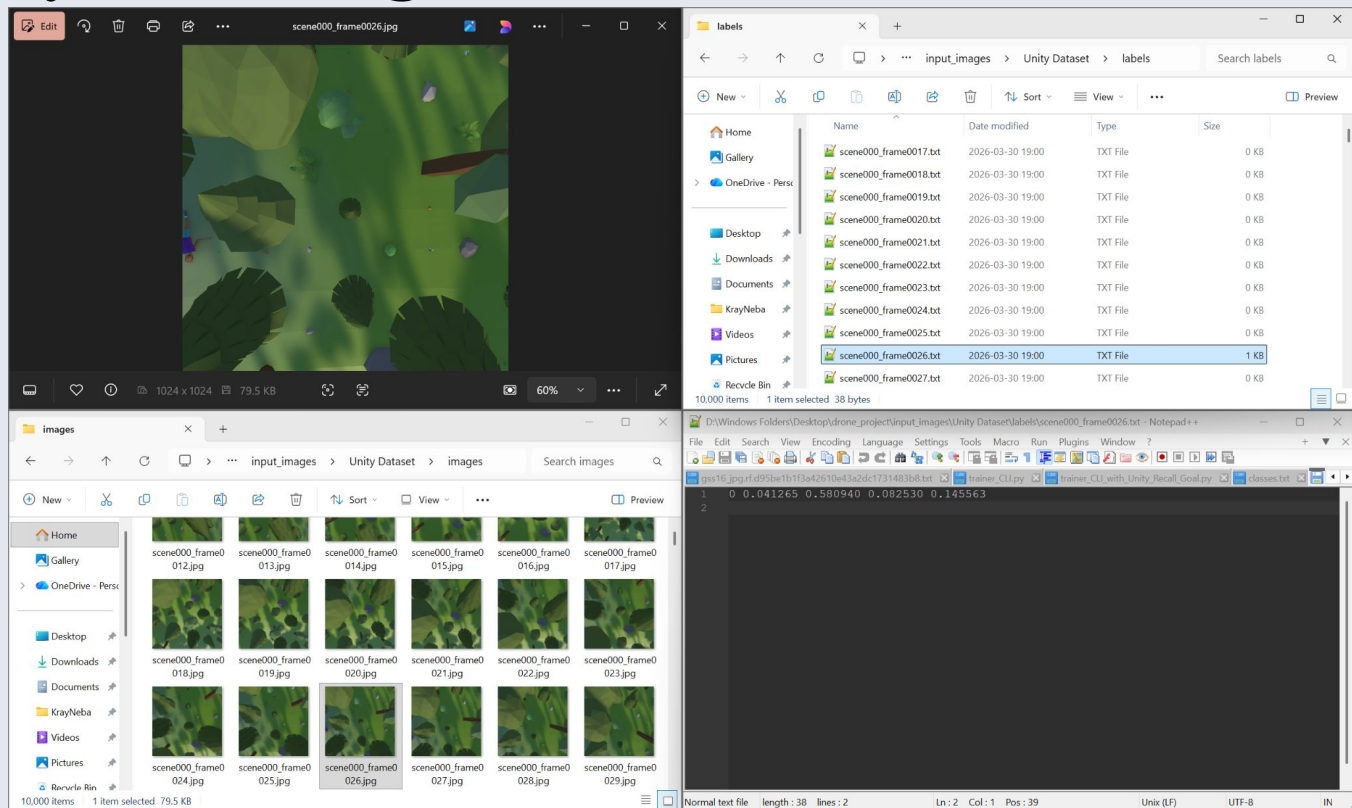
Procedural map generation



Unity Dataset generation



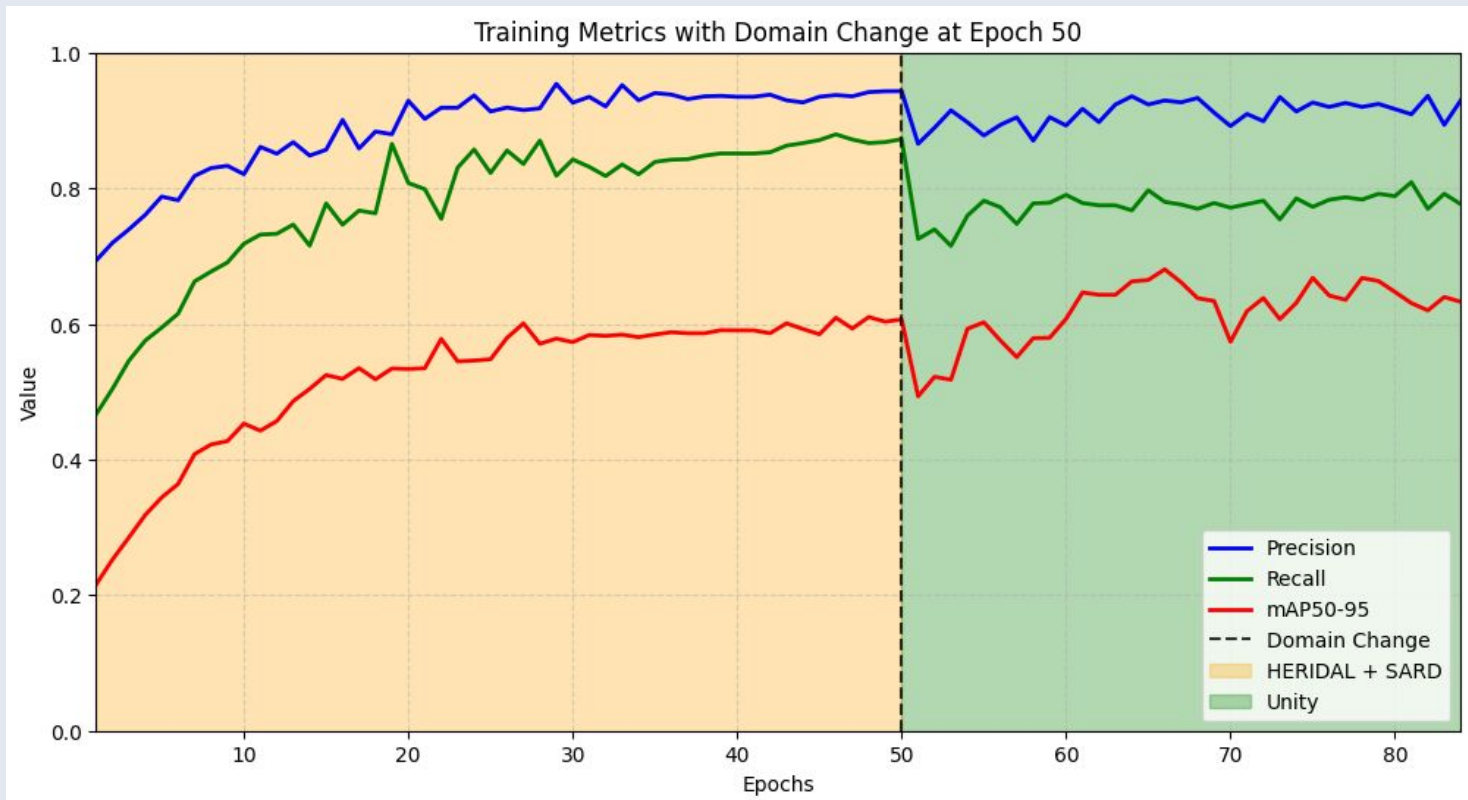
Unity Dataset generation



Domain shift consequences

```
=====
                        BENCHMARK RESULTS
=====
      Model F1 Score Recall Precision      Speed  TP   FP  FN
Unity-FT/last.pt      88.3  82.3%    95.2%  9.4 img/s 320   16  69
production_run_.../last.pt  42.6  44.0%    41.3% 11.4 img/s 171  243 218
      YOLO26M.PT       9.9  64.3%     5.3% 11.0 img/s 250 4430 139
=====
Results saved to: D:\Windows Folders\Desktop\drone_project\brain\benchmark_results.csv
```

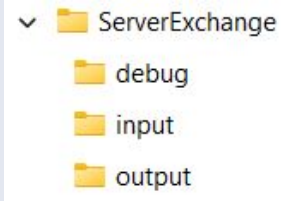
Domain shift consequences



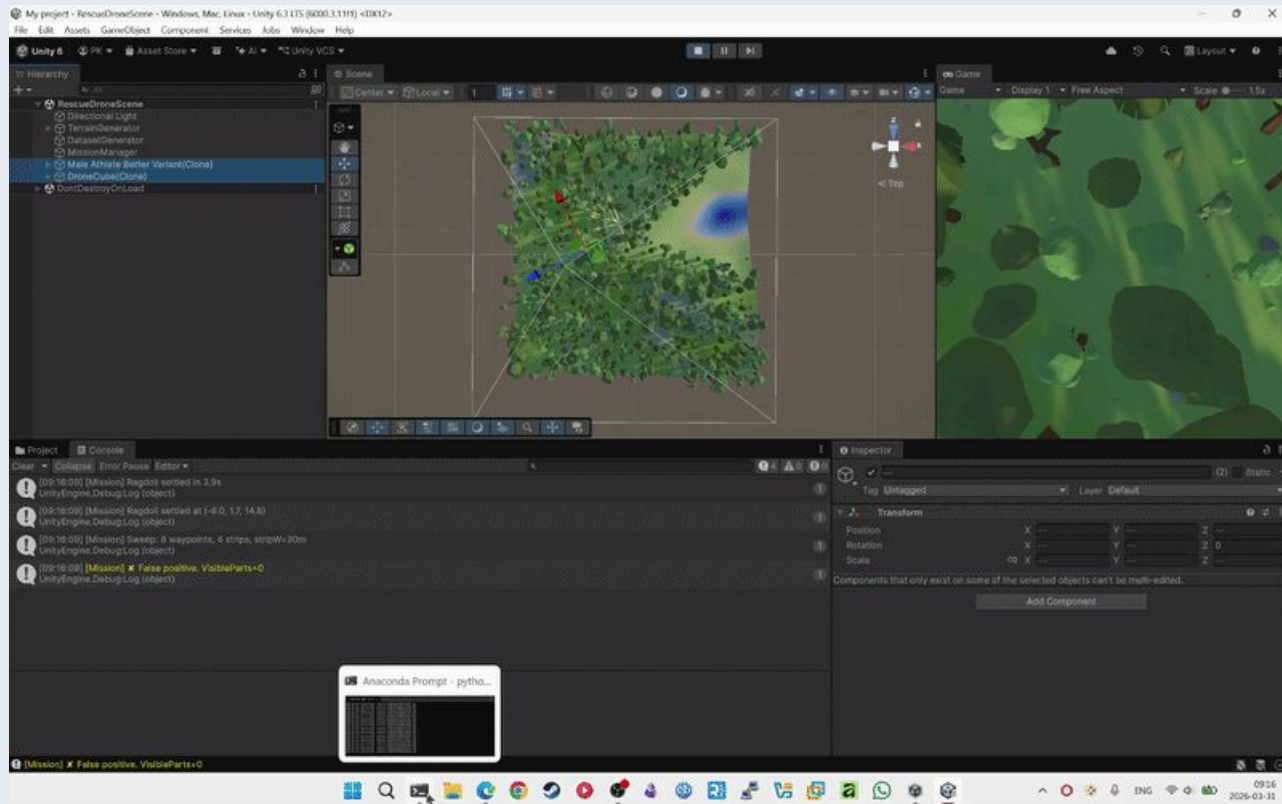
Communication



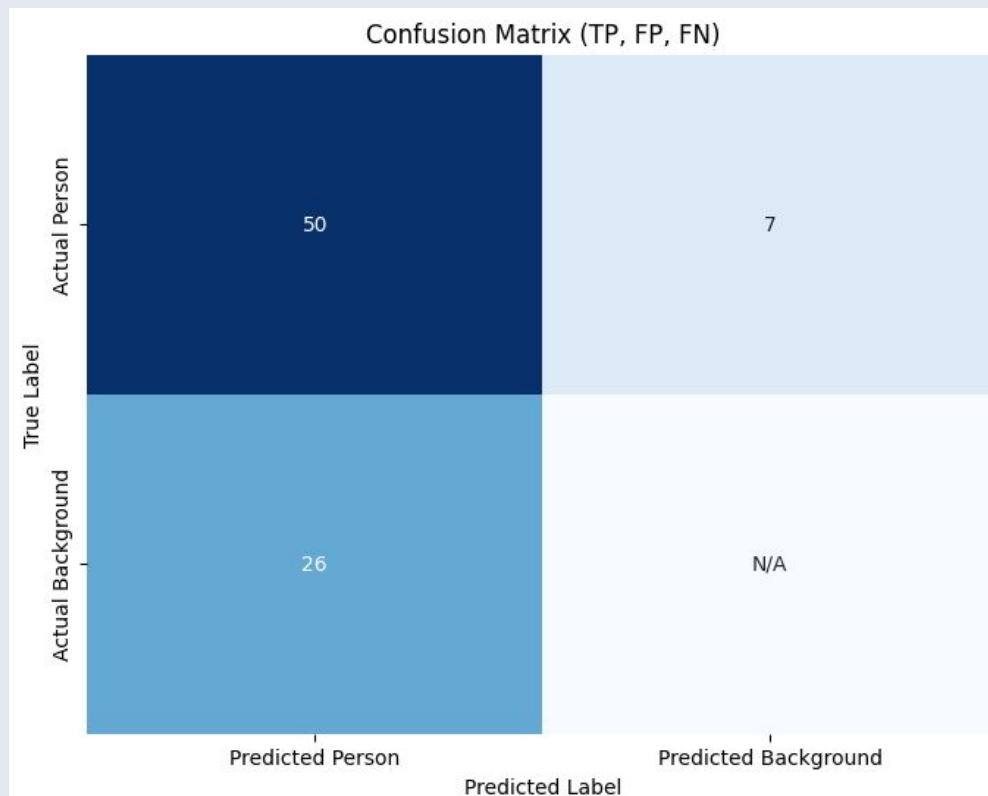
[DroneCamera] sent photo #11: frame_639069162103774920.jpg



“Real” synthetic test

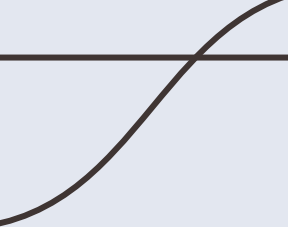


Results for 50 runs on 1 sq. km terrain with threshold = 30%



Ideas to upgrade

- Generate different terrains for the different seasons (like winter, autumn and such)
- Include thermal camera. To try, what is the best
- Include Animals to add complexity
- Do test in real



Thank you for your attention.

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